

17.00 CThN5

Tissues imaging by ultrasonic tagging of transmitted or backscattered photons paths

S. Lévêque-Fort, J. Selb, L. Pottier and A.C. Boccara

Laboratoire d'Optique Physique,
Ecole Supérieure de Physique et Chimie Industrielle, CNRS UPR A0065,
10 rue Vauquelin, F-75008 Paris, France
Phone: 33 1 40 79 45 91 Fax: 33 1 43 36 83 85
fort@optique.espci.fr

Echography is now a routinely used non-invasive technique for imaging through the body. A simple protocol gives quickly an image related to mechanical properties of the tissues. Recently, a complementary technique based on the coupling of light and ultrasound has been developed [1, 2, 3]. This approach allows imaging through several centimeters thick biological tissues with millimeter resolution. The aim of this study is to demonstrate the ability of this imaging system to reveal tumors using *ex-vivo* sample and for different configurations. We also want to emphasize the interest of coupling our set-up with an echography system in order to get both acoustic and acousto-optic signals.

Let us recall the principle of acousto-optic imaging. A single-mode laser at 840 nm illuminates the biological sample. The ultrasonic focused field (focal length 69 mm) at 2.3 MHz induces, in the focal zone, periodic displacements of the scatterers (~ 10 nm) which generates a speckle modulation of the order of the percent. For each position of the transducer focus zone in the sample, we measure the average amplitude of the modulated light. We then reconstruct an image of the sample where the different levels of modulated light reflect the local optical properties of the tissues. When the focus zone is on an absorbing structure, many photons are trapped and so less modulated light reaches the CCD camera. We have obtained 2D and 3D images through biological samples ranging from 2 cm to 3.5 cm which reveal optical absorbing objects. The resolution of the system is determined by the size of the ultrasonic focus zone (2 mm laterally, 25 mm axially), and could be improved by taking a transducer with a greater aperture. Our approach is different from the other teams on several points. Instead of using a single detector which integrates over many speckle grains, we use a CCD camera (256 x 256 pixels) associated to a multichannel lock-in detection. It allows to treat in parallel the modulation of 40000 speckle grains. We then obtain quickly (50 ms) a good statistical value with a signal to noise ratio greater (1% instead of 10^{-4}). The phantoms commonly used are replaced since the beginning by animal tissues or more recently by human tissues presenting a tumor. We start to place the camera CCD and the laser in the same side of the sample for a backscattering configuration, whereas till now only transillumination have been used. It seems to give the same order of modulated signal than in transillumination. We plan to obtain soon an image in this configuration which will open acousto-optic imaging applications on other parts of human body. For the future coupling of acousto-optic imaging with an echograph, we are improving our set-up in terms of signal to noise ratio and time measurement. With this new set-up, optical and acoustical contrast will then be obtained during the same examination and will help doctors for earlier diagnostics.

References

1. L. Wang and X. Zhao, "Ultrasound-modulated optical tomography of absorbing objects buried in dense tissue-simulating turbid media", *Appl. Opt.* **36**, 7277 (1997).
2. M. Kempe, M. Larionov, D. Zaslavsky and A. Z. Cosack, "Acousto-optic tomography with multiply scattered light", *J. Opt. Soc. Am. A*, **14**, 1151 (1997).
3. S. Lévêque, A.C. Boccara, M. Lohse, and H. Saint-Jalmes, "Ultrasound tagging of photons paths in scattering media: parallel speckle modulation processing", *Opt. Lett.* **24**, 181 (1999).

17.15 CThN6

Optical Diffusion Imaging Using Nonlinear Multigrid Optimization

K. J. Webb, J. C. Ye, C. A. Bouman, and R. P. Millane

School of Electrical and Computer Engineering, Purdue University, West
Lafayette, Indiana 47907-1385, USA
Ph: 1-765-454-3573 (Webb), Fax: 1-765-454-5705, webb@ecn.purdue.edu

The measurement situation we consider involves multiple source and detector locations around the perimeter of a highly scattering domain. A frequency domain measurement involves modulating each light source with a CW signal and performing a coherent detection at this modulation frequency. The multiple scatter and absorption within the domain results in a spatially-dependent magnitude and phase of the light intensity (the measurement data). This measured data then must be inverted to form an image based on the spatially varying absorption and scattering parameters in a diffusion equation model [1]. With high scatter and relatively low loss, the diffusion equation provides an accurate forward model. Forming an image involves inverting the diffusion equation to obtain the spatially-dependent scattering and absorption coefficient. This nonlinear inverse problem requires an iterative solution to minimize an appropriate cost function [1]. We have presented a physically-based Bayesian formulation (where instrument shot noise is incorporated) and an iterative coordinate descent (ICD) optimization for the optical diffusion imaging problem [2]. In particular, we have proposed an ICD/Born method which provides high quality reconstructions and is computationally efficient when compared to the conventional iterative Born approximation methods. However, the computational complexity of ICD/Born is still prohibitive for large three dimensional problems. Perhaps more importantly, local optimization methods such as ICD/Born can become trapped in local minima.

Here we demonstrate multigrid algorithms for nonlinear optical diffusion imaging. Multigrid techniques have a number of important advantages over fixed grid methods for this type of problem. First, the convergence of multigrid algorithms is independent of the size of the image, reducing overall computation. Second, the optimization tends to more robustly achieve the global minimum, resulting in improved reconstruction quality. In our multigrid algorithm, each iteration alternates a Born approximation step with a single cycle of a nonlinear multigrid algorithm. At each resolution of the multigrid algorithm, we use the ICD optimization method. A 2-D reconstruction of size 128×128 illustrates our approach (see Fig. 1). The results show that the multigrid optimization reduces computation by a factor of 20-30 in our simulations, relative to a fixed grid solution, while achieving substantially better reconstruction quality. Tissue imaging requires a 3-D inversion, and we expect that the combination of a Bayesian formulation and a multigrid solution will be important.

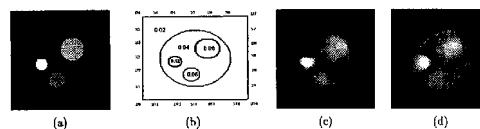


Fig. 1. (a) Gray scale of absorption coefficient for an $8 \times 8 \text{ cm}^2$ tissue model, and (b) contour plot, with units of cm^{-1} , showing positions of sources (SI) and detectors (DI); $\mu_s' = 10^{-3} \text{ cm}^{-1}$. Reconstructions by (c) the fixed resolution ICD/Born algorithm, and (d) the multigrid V-cycle inversion algorithm.

References

1. J. C. Ye, K. J. Webb, R. P. Millane, and T. J. Downar, "Modified distorted Born iterative method with approximate Fréchet derivative for optical diffusion tomography", *J. Opt. Soc. Am. A* **16**, 1814-1826 (1999).
2. J. C. Ye, K. J. Webb, C. A. Bouman, and R. P. Millane, "Optical diffusion tomography using iterative coordinate descent optimization in a Bayesian framework", *J. Opt. Soc. Am. A* **16**, 2400-2412 (1999).